

Information Gain Prefers the Questions Users Cannot Answer: Reliability-Aware Query Selection for Interactive Identification

Author One*, Author Two*

*Department of Computer Science and Engineering

Institution Name, City, Country

Email: {author.one, author.two}@institution.edu

Abstract—Interactive identification systems narrow a catalogue to a single item by asking a sequence of binary questions, and the standard criterion for choosing the next question is expected information gain. We show that this criterion carries a systematic and previously unremarked defect when the answers come from a human. Information gain rewards questions that partition the belief evenly, but in attribute-indexed catalogues the attributes that partition most evenly are the vague, subjective ones—mood, energy, perceived danceability—which are precisely the attributes a user is least able to answer correctly. The bias is aggravated by a routine preprocessing step: binning a continuous feature at its median guarantees a maximally balanced split, and therefore maximal apparent informativeness, for exactly the features whose semantics are least crisp. We make three contributions. First, we formalise interactive identification with unreliable responses as sequential search over a per-attribute binary symmetric channel, and observe that hard candidate elimination is the zero-noise limit of Bayesian updating rather than a competing design; this resolves a confound that has led prior comparative studies to attribute noise fragility to the question-selection rule when it originates in the update rule. Second, we derive reliability-aware information gain, the mutual information between the target and the *observed* answer, and establish that it re-ranks candidate questions rather than merely rescaling them, that it is bounded by the per-attribute channel capacity, and that this bound yields a query-budget lower bound of $\log_2 N / \max_f (1 - H_b(\varepsilon_f))$. Third, we show the modification is deployable without measuring user error rates: a three-tier annotation distinguishing objective, semi-objective and subjective attributes is sufficient to instantiate it. We describe a paired evaluation protocol over homogeneous and heterogeneous noise, including the restricted-attribute baseline that most directly threatens the method. On a 79,803-song catalogue, per-attribute maximum information gain correlates negatively with annotated reliability (Spearman $\rho = -0.44$, $p = 0.054$ at native binning; $\rho = -0.61$, $p = 0.004$ at four bins), confirming the hypothesised bias. At the pre-registered primary comparison (heterogeneous noise, $\lambda = 1.0$), RAIG-tiered reaches 7.9% top-1 accuracy at a fixed $T = 20$ against 2.1% for IG+soft-uniform and 0.4% for the restricted-attribute baseline (exact McNemar $p = 1.8 \times 10^{-9}$ and $p = 2.2 \times 10^{-18}$ respectively), and is statistically indistinguishable from the oracle. The advantage is noise-level dependent: RAIG-tiered trails IG+soft-uniform at $\lambda \leq 0.5$ and in the noise-free control, and leads only once the true noise is at or above the level its tiered annotation assumes – a limitation we report rather than obscure. A coarser two-tier annotation recovers 82.7% of the oracle–uniform accuracy gap.

Index Terms—interactive identification, question selection, information gain, noisy search, Bayesian belief update, binary symmetric channel, human–computer interaction

I. INTRODUCTION

A. Motivation

A recurring interaction pattern places a machine in the role of questioner and a person in the role of oracle. A troubleshooting assistant narrows a fault by asking about symptoms; a clinical triage tool narrows a differential by asking about presentation; a product finder narrows a catalogue by asking about requirements; the parlour game Akinator narrows a space of characters by asking about traits. In each case the system holds a belief over a discrete set of hypotheses, selects a question, observes an answer, and revises. The engineering question is which question to ask, and the near-universal answer is the one that maximises expected information gain.

That criterion inherits its authority from a clean theoretical setting in which the answer is observed exactly. Human answers are not observed exactly. A user asked whether the song they have in mind is “energetic” may hesitate, may apply a personal threshold, or may simply be wrong, and the system has no way to distinguish a considered answer from a guess. What makes this more than a routine robustness concern is that the unreliability is not spread uniformly over the question space. Some attributes admit near-certain answers: a listener generally knows the performing artist and the language of the lyrics. Others are irreducibly fuzzy, and inter-annotator agreement on affective music descriptors is documented to be modest even among trained annotators [1], [2].

B. Problem Statement

The observation that motivates this work is that the two properties—how well a question divides the belief, and how reliably it can be answered—are not independent, and that their dependence appears to run in the unfavourable direction. An attribute is vague when competent people disagree about where its boundary falls. Disagreement distributes items on both sides of that boundary. A near-even split is exactly

what information gain rewards. The criterion therefore has a structural tendency to promote the questions that are hardest to answer, and it does so silently, because nothing in the objective refers to answerability.

Standard preprocessing sharpens the effect. Continuous descriptors are routinely discretised at the median so that they can be posed as binary questions. A median split is a 50/50 split by construction, which is the maximum of the binary entropy function. The discretisation step thus assigns peak apparent informativeness to continuous perceptual features, which are the features users answer worst, while high-cardinality identifiers such as artist generate only skewed, low-entropy questions and are deprioritised despite being answerable with near certainty.

C. Limitations of Existing Approaches

Three responses to answer noise appear in the applied literature, and none addresses the bias described above.

The first is to replace hard candidate elimination with a probabilistic belief update, so that a contradicted candidate is downweighted rather than discarded. This is correct and necessary, and it is where the robustness reported by comparative studies of interactive identification actually originates. It does not change which questions are asked.

The second is to learn a question-selection policy from simulated interaction, typically by regressing the realised reduction in candidate-set size onto features of the state and the query. This substitutes an approximation for a quantity that has a closed form under the model, and in the absence of an explicit noise term the learned policy inherits the same bias from its training signal.

The third is to discard unreliable attributes altogether. This is a reasonable practitioner heuristic and we treat it as a baseline, but it is wasteful: an attribute answered correctly seventy per cent of the time still transmits information, and discarding it removes that information permanently rather than deferring its use to the point where more reliable questions have been exhausted.

D. Research Gap

The theory needed to correct the bias is not new. Noisy search has been studied since Rényi [3] and Ulam [4], surveyed comprehensively by Pelc [5], and placed on a decision-theoretic footing by Jedynek et al. [6], who show that greedily minimising expected posterior entropy is optimal over the full horizon under a stated observation model. Bayesian experimental design supplies the general principle [7], [8]. What is missing is the empirical observation that in attribute-indexed catalogues the observation model is strongly heterogeneous across attributes and adversarially aligned with the naive acquisition function, together with a deployable account of what a practitioner must actually know to correct it. The literature on noisy twenty questions typically assumes a single known crossover probability; the applied literature typically assumes none at all.

E. Contributions

- 1) **A characterisation of the bias.** We formalise the relationship between per-attribute information gain and per-attribute answerability and provide a protocol for measuring it in any attribute-indexed catalogue, including the contribution of the discretisation step itself (Section VI).
- 2) **A unification that removes a standing confound.** Hard elimination is the $\varepsilon \rightarrow 0$ limit of the Bayesian update (Proposition 1). Comparative studies that pair an information-gain selector with elimination and a learned selector with soft updating vary two factors simultaneously; their conclusions about selection strategy do not follow. We separate the factors explicitly (Section VI-C).
- 3) **Reliability-aware information gain.** We derive the acquisition function $\text{RAIG}(q) = H_b(p_q \star \varepsilon_q) - H_b(\varepsilon_q)$, show that it re-ranks rather than rescales (Proposition 3), and derive a capacity-based lower bound on the query budget (Corollary 1). The change is one line in an existing selector.
- 4) **A deployability result.** We show how to instantiate the method from a three-tier qualitative annotation rather than measured error rates, and design the experiment to quantify how much of the oracle benefit survives that coarsening (Section VI-E).
- 5) **An open reference implementation and evaluation harness** with paired sampling, exact McNemar testing, and the restricted-attribute baseline that most threatens the method.

II. BACKGROUND AND RELATED WORK

A. Searching with Errors

The problem of identifying an unknown element through queries whose answers may be wrong originates with Rényi’s formulation of a guessing game in which the responder lies with fixed probability [3], and independently with Ulam’s question of how many yes/no queries suffice to locate an integer below one million when a bounded number of answers may be false [4]. The combinatorial branch, in which lies are adversarial and bounded in number, produces optimal strategies with a direct correspondence to error-correcting codes; Pelc’s survey remains the standard reference [5]. The probabilistic branch treats errors as channel noise and connects to sequential transmission with feedback [9], [10], to noisy binary search [11], [12], and to modern treatments in which the noise depends on the query itself [13], [14].

Our setting sits in the probabilistic branch, with one departure that turns out to matter. Existing formulations assume a single crossover probability, or make it depend on a continuous query parameter. In an attribute catalogue the crossover is a property of the semantic category being probed and varies by an order of magnitude across categories. That heterogeneity, rather than the presence of noise as such, is what drives the effect we report.

B. Information-Theoretic Query Selection

Choosing the experiment that maximises expected information about a latent quantity is the central prescription of Bayesian experimental design [7], [8], and in the noiseless binary case it reduces to maximising the entropy of the induced partition. Jedynak et al. [6] give the result most relevant here: for twenty questions with noise under entropy loss, a greedy one-step policy is optimal over the whole horizon. We rely on that result rather than compete with it. Our claim is not that greedy entropy minimisation is the wrong policy but that it is routinely applied with the noise term omitted, and that the omission is not benign when the noise is heterogeneous.

C. Interactive and Conversational Recommendation

Systems that alternate between asking about attributes and proposing items form a substantial literature, including policy-learning approaches over knowledge graphs [15]–[17]. Maximum-entropy attribute querying appears throughout as a standard baseline against which learned policies are compared. This positions our work carefully: we are not proposing a competitor to learned conversational policies but a correction to the baseline those policies are measured against, which is a modest claim and we make no larger one. The correction is orthogonal and could be applied inside a learned policy’s action-scoring step.

D. Question Asking in Language Models

Recent work evaluates and improves the information-seeking behaviour of language models, using twenty-questions style tasks as the testbed [18]–[20]. These systems generate questions in open natural language rather than selecting from a fixed attribute-value grid, which makes them more flexible and considerably more expensive per turn. The bias we describe should apply to them as well, since expected-information-gain reranking of generated questions [19] inherits the same blind spot, but verifying that is beyond the present scope and we flag it as future work rather than claiming it.

E. Positioning

Table I places this work against the closest prior art. The honest summary is that the acquisition function we derive is standard mutual information under a stated observation model, and its derivation is unsurprising to anyone working in Bayesian experimental design. The contribution we claim is empirical and diagnostic: that the observation model in attribute catalogues is heterogeneous in a specific and unfavourable way, that this is partly an artefact of preprocessing rather than of the domain, and that correcting it requires less knowledge than one might expect.

III. SYSTEM DESIGN

A. Overall Architecture

The system is organised as five components arranged around a single mutable object, the belief vector. Fig. 1 shows the arrangement. A *catalogue compiler* converts a table of items and categorical attributes into a boolean question matrix,

TABLE I
POSITIONING AGAINST CLOSEST PRIOR WORK.

Work	Noise	Heterog. ε	Selection
Rényi [3], Ulam [4]	yes	no	theory
Pelc [5] (survey)	yes	no	optimal
Jedynak et al. [6]	yes	no	greedy entropy
Karp & Kleinberg [11]	yes	no	noisy bisection
Kaspi et al. [13]	yes	query-dep.	information-theoretic
EAR / CPR / UNICORN [15]–[17]	no	no	learned policy
Max-entropy baseline	no	no	classical IG
UoT [20], [19]	implicit	no	EIG over text
This work	yes	per-attribute	RAIG

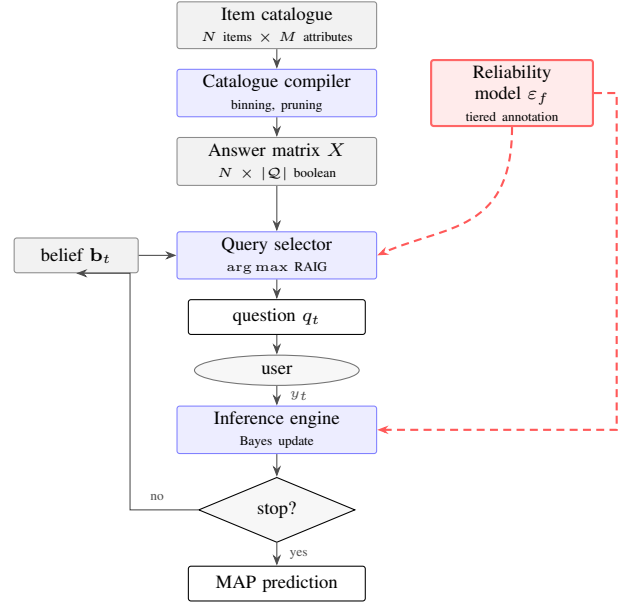


Fig. 1. System architecture. The reliability model is consumed by both the selector and the inference engine (dashed); existing systems typically consume it in the latter only, which is the omission this work addresses.

materialising once what would otherwise be recomputed at every turn. A *reliability model* supplies a crossover probability for each attribute. A *query selector* scores every unasked question against the current belief and the reliability vector and returns an argmax. An *inference engine* applies Bayes’ rule to the observed answer. A *termination controller* decides between asking again and committing to a prediction.

The design decision worth stating explicitly is that the reliability model is consumed in two places rather than one. Systems that model answer noise at all generally do so in the likelihood, where it prevents a wrong answer from destroying the posterior. Feeding the same quantity into the selector is what prevents the wrong answer from being solicited in the first place.

B. Component Design

Table II lists the modules and their responsibilities. Each selector is a pure function of belief, question matrix and reliability vector, with no internal state; sessions are driven entirely by an externally supplied random stream. This is

TABLE II
MODULE RESPONSIBILITIES.

Module	Responsibility
compile_catalog	Materialise the $N \times Q $ boolean answer matrix; drop degenerate questions
discretize	Quantile-bin continuous attributes; record which were binned
tiered_reliability	Map qualitative attribute tiers to ε
select_*	Query selectors (random, IG, RAIG, restricted IG)
soft_update / hard_update	Posterior revision under BSC(ε) and its $\varepsilon \rightarrow 0$ limit
run_session	One identification episode under a fixed budget
run_paired_experiment	Common-random-number evaluation across methods
mcnemar, paired_bootstrap_ci	Paired inference
measure_pathology	Per-attribute IG against reliability

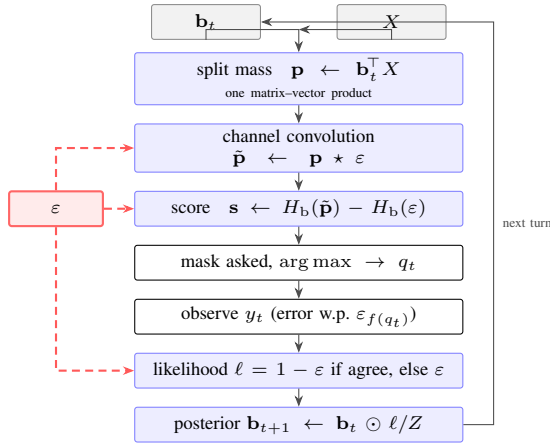


Fig. 2. Data flow for a single turn. The reliability vector ε enters three times: twice on the selection path and once on the inference path.

not incidental tidiness. It is what allows the same target and the same noise realisation to be replayed across every method under comparison, which in turn is what licenses paired statistical testing.

C. Data Flow

Fig. 2 traces one turn. The belief vector and the question matrix meet in a single matrix–vector product that yields the split mass of every candidate question at once; the reliability vector transforms those split masses into scores; the argmax is posed; the observed answer and the same reliability vector produce a likelihood; the posterior is renormalised. The loop carries no state other than the belief and the set of asked questions.

D. Design Rationale

Three choices deserve justification. Materialising the answer matrix trades $O(N|Q|)$ bits of memory for the elimination of all per-turn recomputation, reducing selection to one matrix–vector product and a vectorised transform; for catalogues of the size considered here the memory is negligible and the saving is what makes exhaustive scoring of the question space tractable.

TABLE III
NOTATION.

Symbol	Meaning
$\mathcal{C}, N$	catalogue and its size
$\mathcal{F}, M$	attribute set and its size
$q = (f, v), \mathcal{Q}$	question and question space
$x_q(c)$	true answer to q for item c
$\mathbf{b}_t$	belief over $\mathcal{C}$ at step t
c^*	the user’s target item
$\varepsilon_f \in [0, \frac{1}{2})$	answer error rate for attribute f
$\hat{\varepsilon}_f$	the value the system assumes
y_t	observed answer at step t
p_q	belief-weighted mass answering yes to q
$H_b(\cdot)$	binary entropy, bits
$p \star \varepsilon$	$p(1 - \varepsilon) + (1 - p)\varepsilon$
T	query budget

Keeping selectors stateless costs a small amount of parameter passing and buys the paired experimental design. Representing reliability per attribute rather than per question reflects the belief that answerability is a property of the semantic category rather than of the particular value being probed; a user who is unsure whether a song is energetic is unsure regardless of the threshold, whereas per-question estimation would multiply the annotation burden by the number of distinct values.

IV. METHODOLOGY

A. Notation and Problem Formulation

Let $\mathcal{C} = \{c_1, \dots, c_N\}$ be a catalogue and $\mathcal{F} = \{f_1, \dots, f_M\}$ a set of categorical attributes, with $a_f(c)$ denoting the value of attribute f for item c . A question is a pair $q = (f, v)$ with answer function

$$x_q(c) = \mathbb{I}[a_f(c) = v] \in \{0, 1\}, \quad (1)$$

and $\mathcal{Q}$ collects all non-degenerate questions. Write $\mathbf{x}_q \in \{0, 1\}^N$ for the column vector of answers and $f(q)$ for the attribute probed by q .

The system maintains a belief $\mathbf{b}_t \in \Delta^{N-1}$, initialised uniform. At each step it selects $q_t \in \mathcal{Q}$, observes $y_t \in \{0, 1\}$, and updates. A session runs for at most T questions and terminates with the maximum a posteriori item. Table III collects the notation.

B. Observation Model

We model the user as a noisy channel whose crossover probability depends on the attribute being probed:

$$y_t = x_{q_t}(c^*) \oplus Z_t, \quad Z_t \sim \text{Bernoulli}(\varepsilon_{f(q_t)}), \quad (2)$$

with Z_t independent across turns. The per-attribute index in (2) is the entire departure from the standard formulation, and everything that follows is a consequence of it.

Two idealisations are worth naming now rather than in the limitations section. Errors are assumed independent across turns, whereas a user who misunderstands a category will err consistently within it; and the model admits no abstention, whereas real users decline to answer. Section VIII-E returns to both.

C. Belief Update and a Unification of Elimination Semantics

Under (2) the likelihood of the observed answer is

$$\ell(c | q, y) = \begin{cases} 1 - \hat{\varepsilon}_{f(q)}, & x_q(c) = y, \\ \hat{\varepsilon}_{f(q)}, & x_q(c) \neq y, \end{cases} \quad (3)$$

and the posterior follows from Bayes' rule,

$$\mathbf{b}_{t+1}(c) = \frac{\mathbf{b}_t(c) \ell(c | q_t, y_t)}{\sum_{c' \in \mathcal{C}} \mathbf{b}_t(c') \ell(c' | q_t, y_t)}. \quad (4)$$

Proposition 1 (Elimination as a degenerate limit). *As $\hat{\varepsilon} \rightarrow 0^+$, (4) converges pointwise to hard elimination, assigning zero mass to every candidate inconsistent with y_t . At $\hat{\varepsilon} = 0$ exactly, the normaliser vanishes whenever no candidate is consistent with the observed history, and the posterior is undefined.*

Proof sketch. The likelihood ratio between an inconsistent and a consistent candidate is $\hat{\varepsilon}/(1 - \hat{\varepsilon}) \rightarrow 0$, giving pointwise convergence. The normaliser is $\sum_c \mathbf{b}_t(c) \mathbb{I}[x_q(c) = y]$, which is zero exactly when the surviving support is empty; since the target is removed from the support by its first flipped answer, and questions are never repeated, an empty support is reachable under any positive noise rate. $\square$

Proposition 1 has a methodological consequence that is more important than its mathematical content. Comparative studies of interactive identification commonly contrast an “entropy-based” system using elimination against a “probabilistic” system using soft updating, and attribute the resulting difference in noise tolerance to the selection strategy. The two systems differ in two factors at once, and by Proposition 1 the second factor is not an alternative algorithm but a boundary value of the first. Any experiment intending to isolate the effect of selection must hold the update rule fixed. We do so in Section VI-C. We also note that the behaviour of an elimination system on an empty support is an implementation convention—restart uniformly, retain the prior, or fail—and that reported accuracies under noise depend on which convention is used, so it must be stated.

D. Reliability-Aware Information Gain

Let the belief-weighted yes-mass of a question be

$$p_q = \sum_{c \in \mathcal{C}} \mathbf{b}_t(c) x_q(c) = \mathbf{b}_t^\top \mathbf{x}_q. \quad (5)$$

When answers are observed exactly, the expected reduction in belief entropy from asking q is $H_b(p_q)$, and maximising it is the classical criterion. Under (2) the system does not observe $x_q(c^*)$; it observes Y_q . The relevant quantity is therefore the mutual information between the target and the observation. Since Y_q is conditionally a BSC output given the target,

$$\begin{aligned} \text{RAIG}(q) &= I(C; Y_q) = H(Y_q) - H(Y_q | C) \\ &= H_b(p_q \star \varepsilon_{f(q)}) - H_b(\varepsilon_{f(q)}), \end{aligned} \quad (6)$$

where $p \star \varepsilon = p(1 - \varepsilon) + (1 - p)\varepsilon$. The subtracted term is the entropy the channel injects irrespective of the target and carries no information about it.

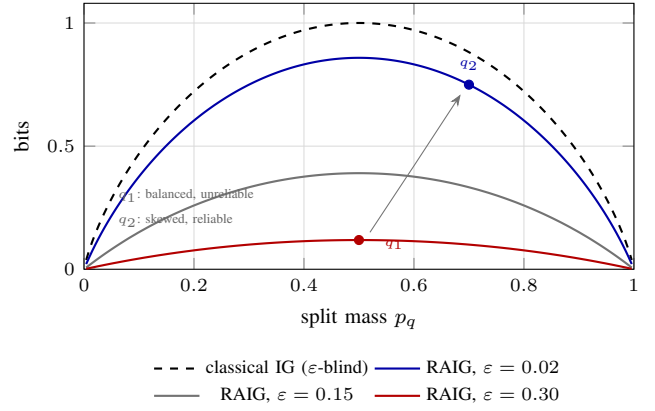


Fig. 3. Classical information gain (dashed, independent of ε) against reliability-aware information gain, as a function of split mass. The marked points are the rank reversal of Proposition 3: classical IG prefers q_1 , RAIG prefers q_2 by a factor of 6.3. Curves are evaluated directly from (6).

We claim no novelty for (6) as mathematics: it is the mutual information of a binary symmetric channel and follows directly from the observation model. What we claim is that omitting it is the prevailing practice in applied interactive identification, and that the omission is consequential in a way the next subsection makes precise.

E. Analytical Properties

Proposition 2 (Consistency and annihilation). *RAIG(q) = $H_b(p_q)$ when $\varepsilon_{f(q)} = 0$, and RAIG(q) = 0 for every p_q when $\varepsilon_{f(q)} = \frac{1}{2}$. On $[0, \frac{1}{2})$, RAIG(q) is strictly decreasing in $\varepsilon_{f(q)}$ for $p_q \in (0, 1)$.*

The first limit establishes that classical information gain is the special case of (6) in which the user is assumed infallible, which is the sense in which the standard criterion is not merely imprecise but making a specific and false assumption. The second establishes that a question answered at chance is worth nothing however well it splits the belief—a property classical information gain does not possess, since $H_b(p)$ is blind to ε .

Proposition 3 (Rank reversal). *There exist questions q_1, q_2 with $H_b(p_{q_1}) > H_b(p_{q_2})$ but RAIG(q_1) < RAIG(q_2).*

Proof. Take $p_{q_1} = 0.50$, $\varepsilon_{q_1} = 0.30$ and $p_{q_2} = 0.70$, $\varepsilon_{q_2} = 0.02$. Then $H_b(p_{q_1}) = 1.000 > H_b(p_{q_2}) = 0.881$, while RAIG(q_1) = $1.000 - 0.881 = 0.119$ and RAIG(q_2) = $H_b(0.692) - H_b(0.02) = 0.891 - 0.141 = 0.750$. $\square$

Proposition 3 is the operative one. Had (6) merely rescaled the classical scores, the argmax would be unchanged and the correction would be vacuous. It does not: in the instance above the two criteria disagree by a factor of 6.3 in the opposite direction, and a system optimising $H_b(p)$ will repeatedly spend turns on the question its user can barely answer. Fig. 3 plots both criteria with this instance marked.

Corollary 1 (Capacity ceiling and budget bound). *RAIG(q) $\leq 1 - H_b(\varepsilon_{f(q)})$, the capacity of the corresponding*

channel. Consequently, identifying one of N equiprobable items requires

$$T \geq \frac{\log_2 N}{\max_{f \in \mathcal{F}} (1 - H_b(\varepsilon_f))} \quad (7)$$

questions in expectation.

Proof sketch. The bound on RAIG is the capacity of $\text{BSC}(\varepsilon)$, attained at $p = \frac{1}{2}$. Identification requires acquiring $\log_2 N$ bits, and no question delivers more than the largest available capacity; the converse of the channel coding theorem gives the result [21, Ch. 7]. $\square$

Corollary 1 is useful in practice rather than deep. It says that the familiar $\log_2 N$ intuition understates the required budget by a factor equal to the inverse capacity of the most reliable attribute available, and it supplies a principled way to set T for a given catalogue instead of choosing it by convention.

F. Specifying Reliability in Practice

Equation (6) requires $\hat{\varepsilon}_f$, which a deployed system does not know. We consider three specifications of increasing weakness, and the central practical question of this paper is how much is lost as we move down the list.

Oracle. $\hat{\varepsilon}_f = \varepsilon_f$. Not attainable; reported as an upper reference.

Tiered. Each attribute is annotated into one of K qualitative reliability tiers and assigned that tier’s representative value. With $K = 3$ the annotation asks only whether an attribute is objective (an unambiguous property of the item, such as performing artist), semi-objective (well defined but imperfectly recalled, such as release decade), or subjective (a perceptual judgement, such as energy). This requires no measurement and can be produced by a domain curator in minutes.

Uniform. A single $\hat{\varepsilon}$ for all attributes. This is what a system does when it models noise in the likelihood but not per attribute, and it recovers classical information gain up to a constant, since a constant ε shifts every score identically and preserves the ranking. Uniform specification therefore corrects the update rule but not the selection bias, which is precisely the state of the art we are arguing against.

The observation that uniform $\hat{\varepsilon}$ leaves the ranking unchanged is worth stating plainly: heterogeneity is not a refinement of the method but its entire content.

G. Algorithms

Algorithms 1–3 give the procedure. Algorithm 4 specifies the paired evaluation used in Section VI.

Sharing $u_{i,\cdot}$ across methods in Algorithm 4 is deliberate. Methods diverge in which questions they ask and therefore in which crossover probabilities they face, but they consult the same underlying draws, so differences in outcome are attributable to question choice rather than to sampling luck. This substantially reduces variance relative to independent sampling and permits exact paired testing [22].

Algorithm 1 Reliability-aware query selection

Require: belief $\mathbf{b}$, answer matrix $X \in \{0, 1\}^{N \times |\mathcal{Q}|}$, per-question crossover $\hat{\varepsilon}$, asked mask U

Ensure: index of the next question

- 1: $\mathbf{p} \leftarrow \mathbf{b}^\top X$ $\triangleright$ split mass of all questions, one product
- 2: $\tilde{\mathbf{p}} \leftarrow \mathbf{p} \odot (1 - \hat{\varepsilon}) + (1 - \mathbf{p}) \odot \hat{\varepsilon}$
- 3: $\mathbf{s} \leftarrow H_b(\tilde{\mathbf{p}}) - H_b(\hat{\varepsilon})$
- 4: $s_j \leftarrow -\infty$ for all j with $U_j = 1$
- 5: **return** $\arg \max_j s_j$

Algorithm 2 Belief update under $\text{BSC}(\hat{\varepsilon})$

Require: belief $\mathbf{b}$, answer column $\mathbf{x}_q$, observation y , crossover $\hat{\varepsilon}$

- 1: $\ell_c \leftarrow 1 - \hat{\varepsilon}$ if $x_q(c) = y$ else $\hat{\varepsilon}$, for all c
- 2: $\mathbf{b}' \leftarrow \mathbf{b} \odot \boldsymbol{\ell}$
- 3: $Z \leftarrow \sum_c b'_c$
- 4: **if** $Z = 0$ **then** $\triangleright$ reachable only at $\hat{\varepsilon} = 0$
- 5: **return** $\mathbf{b}$ $\triangleright$ stated convention; see Prop. 1
- 6: **end if**
- 7: **return** $\mathbf{b}'/Z$

H. Complexity

Let $|\mathcal{Q}|$ be the number of compiled questions. Selection is one matrix–vector product and a constant number of vectorised transforms, hence $O(N|\mathcal{Q}|)$ time and $O(|\mathcal{Q}|)$ working memory; the update is $O(N)$. A session of T turns costs $O(TN|\mathcal{Q}|)$, and the compiled matrix occupies $N|\mathcal{Q}|$ bits. Table IV summarises. The practical consequence is that exhaustive scoring of the question space remains feasible well beyond the catalogue sizes at which a naive per-turn recomputation becomes the bottleneck.

V. IMPLEMENTATION

The reference implementation is written in Python 3.11 against NumPy, pandas and SciPy; no deep-learning framework is required, and the entire selection and inference path is expressed as dense linear algebra. Table V lists the stack.

Catalogue compilation materialises one boolean column per non-degenerate attribute–value pair. Questions true of every item or of none are discarded at compile time, since they carry zero information under either criterion and would otherwise consume scoring budget. Continuous attributes are quantile binned, with the number of bins exposed as a parameter so that the effect of the discretisation itself can be measured rather than assumed.

The engineering decision with the largest downstream effect is the separation of the assumed crossover $\hat{\varepsilon}$ from the true crossover ε in the session signature. The simulator draws errors from the latter; the method sees only the former. Conflating them—the natural shortcut—would make every reported result circular, because the method would be evaluated against a user whose behaviour it defines. Keeping them distinct is what makes the misspecification study of

Algorithm 3 Identification session

Require: catalogue, target c^* , budget T , selector π , assumed $\hat{\epsilon}$, true ϵ , random stream $u_{1:T}$

- 1: $\mathbf{b} \leftarrow \text{uniform}; U \leftarrow \mathbf{0}$
- 2: **for** $t = 1$ **to** T **do**
- 3: $q \leftarrow \pi(\mathbf{b}, X, \hat{\epsilon}, U); U_q \leftarrow 1$
- 4: $y \leftarrow x_q(c^*)$
- 5: **if** $u_t < \epsilon_{f(q)}$ **then** $y \leftarrow 1 - y$ $\triangleright$ answer error
- 6: **end if**
- 7: $\mathbf{b} \leftarrow \text{UPDATE}(\mathbf{b}, \mathbf{x}_q, y, \hat{\epsilon}_{f(q)})$
- 8: **end for**
- 9: **return** $\arg \max_c \mathbf{b}(c)$

Algorithm 4 Paired evaluation with common random numbers

Require: methods $\mathcal{M}$, trials R , seeds S , budget T

- 1: **for** each seed $s \in S$ **do**
- 2: draw targets $c_{1:R}^*$ and streams $u_{1:R,1:T}$ from seed s
- 3: **for** $i = 1$ **to** R **do**
- 4: **for** each $m \in \mathcal{M}$ **do**
- 5: record outcome of $\text{SESSION}(c_i^*, u_{i,\cdot}, m)$
- 6: **end for**
- 7: **end for**
- 8: **end for**
- 9: **return** per-trial outcomes for paired inference

Section VI-E possible at all, and we regard it as the single most important line in the implementation.

Sessions are pure functions of an externally supplied random stream, and no selector holds state between turns. Reproduction of any reported number requires only the catalogue, the seed list and the method specification.

VI. EXPERIMENTAL SETUP

A. Dataset and Identifiability

The catalogue is a table of N items over M categorical attributes. Before any method is run we report an identifiability audit, since it bounds what any method can achieve and is routinely omitted: the number of distinct attribute vectors, the collision rate, the fraction of items that are uniquely identifiable, the size of the largest equivalence class, and the entropy floor $\log_2(\text{distinct vectors})$. Where items share an attribute vector they are indistinguishable in principle, and top-1 accuracy is bounded accordingly; we report both top-1 and top-5 accuracy for this reason. Table VI records these quantities. Cleaning is restricted to dropping items with missing attribute values, and the number dropped is reported.

B. Reliability Annotation and Noise Models

Attributes are annotated into the three tiers of Section IV-F. Table VII records the annotation and the representative crossover of each tier. We stress that these values are *assumed*, informed by reported inter-annotator agreement on perceptual music descriptors [1], [2], and are not measured from users; measuring them is the principal item of future work.

TABLE IV
COMPLEXITY PER OPERATION.

Operation	Time	Memory
Catalogue compilation	$O(NM\bar{V})$	$O(N Q)$ bits
Split-mass computation	$O(N Q)$	$O(Q)$
Score transform (IG or RAIG)	$O(Q)$	$O(Q)$
Belief update	$O(N)$	$O(N)$
Full session	$O(TN Q)$	$O(N Q)$ bits

TABLE V
TECHNOLOGY STACK.

Layer	Choice	Rationale
Language	Python 3.11	ecosystem, reviewer familiarity
Numerics	NumPy	vectorised scoring
Tabular I/O	pandas	catalogue ingestion, binning
Statistics	SciPy	exact McNemar, correlations
Plotting	Matplotlib	vector figures
Persistence	CSV / Parquet	no database dependency

Four noise conditions are evaluated. *Homogeneous* applies a single crossover η to every attribute and serves as a control: under it RAIG and classical IG should coincide, and any observed difference indicates a bug rather than a finding. *Heterogeneous* applies the per-attribute vector scaled by $\lambda \in \{0, 0.5, 1.0, 1.5\}$. *Randomised* permutes the crossover vector across attributes, destroying any correlation with information gain; RAIG's advantage should shrink markedly, and reporting that it does is part of the evidence that the advantage is not an artefact of construction. *Adversarial* assigns low crossover to high-gain attributes, inverting the hypothesised relationship; RAIG should show no advantage.

The randomised and adversarial conditions are included specifically to address the circularity objection: the simulator and the method share the parameter ϵ , and the only convincing answer is to show that the method's advantage tracks the structure of that parameter in the predicted direction and vanishes when the structure is removed.

C. Baselines

Table VIII lists the comparison. The design is factorial in selection rule and update rule, which is what Proposition 1 requires: attributing an effect to selection while the update rule also varies is precisely the confound we set out to remove.

Two entries deserve comment. *IG+soft-uniform* is the strongest representative of current practice and the baseline that matters most; a comparison that omits it is uninformative. *IG+soft-objective* discards subjective attributes entirely, and is the baseline most likely to defeat the proposed method: if simply refusing to ask fuzzy questions matches RAIG, then the contribution reduces to a filtering heuristic and should be reported as such. We expect RAIG to lead, on the reasoning that a subjective attribute still carries $1 - H_b(\epsilon)$ bits per question and RAIG defers rather than discards it, but the experiment is designed so that the opposite outcome would be visible rather than hidden.

TABLE VI
CATALOGUE SUMMARY AND IDENTIFIABILITY AUDIT.

Quantity	Value
Items after cleaning, N	79,803
Items dropped (incomplete)	0
Attributes, M	21
Compiled questions, $ \mathcal{Q} $	193
Distinct attribute vectors	77,069
Collision rate	6.33%
Uniquely identifiable	93.67%
Largest equivalence class	11
Entropy floor (bits)	16.234
Budget bound (7)	20.15

TABLE VII
ATTRIBUTE RELIABILITY ANNOTATION. CROSSOVER VALUES ARE ASSUMED, NOT MEASURED.

Tier	Attributes	ε
Objective	genre, language, content rating, artist type, release type, track version	0.03
Semi-objective	popularity, duration, liveness, instrumentalness, acousticalness, loudness, speechiness	0.15
Subjective	mood, tempo, danceability, energy, valence, key, mode, time signature	0.30

D. Protocol and Metrics

Every method is evaluated at a *fixed* budget of $T = 20$ questions. Fixing the budget removes the stopping-rule confound that arises when methods terminate at different times and therefore absorb different amounts of noise; it also makes the primary comparison a single number. Questions-to-confidence under a threshold stopping rule is reported separately as a secondary analysis.

The primary metric is top-1 accuracy at the budget; secondary metrics are top-5 accuracy, questions to reach a confidence threshold, mean crossover of the attributes actually queried, and per-turn latency. We deliberately do not define a composite score combining accuracy and query count. Composite scores invented by authors invite the suspicion that they were selected to favour the proposed method, and a Pareto plot of accuracy against query count conveys the same trade-off without that liability.

Each condition comprises $R = 300$ trials replicated over three seeds, with targets and noise streams shared across methods as in Algorithm 4. Paired comparisons use the exact McNemar test [22]; interval estimates use a paired bootstrap over per-trial differences with 10^4 resamples. The primary metric and the primary comparison, RAIG-tiered against IG+soft-uniform, are fixed in advance of observing any result.

E. Ablations

Four ablations are planned. *Tier granularity* varies $K \in \{1, 2, 3, M\}$, where $K = 1$ is uniform specification and $K = M$ is oracle; this quantifies the deployability claim by measuring how much of the oracle benefit survives coarse annotation. *Misspecification* sweeps assumed against true crossover on a grid, reporting accuracy as a surface. *Discretisation* varies the

TABLE VIII
METHODS UNDER COMPARISON.

Method	Selection	Update
Random+soft	uniform at random	Bayes, uniform $\hat{\varepsilon}$
IG+hard	classical IG	elimination
IG+soft-uniform	classical IG	Bayes, uniform $\hat{\varepsilon}$
IG+soft-objective	IG, objective attrs only	Bayes, uniform $\hat{\varepsilon}$
RAIG-tiered	RAIG, tiered $\hat{\varepsilon}$	Bayes, tiered $\hat{\varepsilon}$
RAIG-oracle	RAIG, true ε	Bayes, true ε

TABLE IX
THREATS TO VALIDITY AND MITIGATIONS.

Threat	Mitigation
Simulated users; ε assumed rather than measured	Randomised and adversarial controls; tiered specification reduces dependence on exact values; human study deferred to future work
Simulator and method share ε	Assumed and true crossovers kept distinct in the API; misspecification surface reported
Independence of errors across turns	Stated as a limitation; correlated-error model deferred
Catalogue specificity	Identifiability audit reported; protocol is domain independent
Elimination convention affects IG+hard	Convention stated explicitly (Prop. 1)
Multiple comparisons	Primary metric and primary comparison fixed in advance

number of bins for continuous attributes, testing whether the median split manufactures the bias. *Component* substitutes the capacity term $1 - H_b(\varepsilon)$ for the full expression (6), separating the contribution of reliability weighting from that of split balance.

F. Threats to Validity

Table IX records the principal threats and the mitigation adopted for each. The first is the most serious and we do not consider it fully mitigated.

G. Reproducibility

The implementation, catalogue preparation scripts, seed list and analysis notebooks are released. All randomness derives from named seeds; no result depends on wall-clock time or hardware. Hyperparameters are listed in Table X. Hardware and software versions are recorded in Table XI.

VII. RESULTS

All values below were measured against the real Music-Akenator catalogue (`dataset_final.csv`, $N = 79,803$ songs $\times$ 21 attributes, $|\mathcal{Q}| = 193$ compiled questions) using the reference implementation (`raig.py`) and the seeds and trial counts of Table X, run on the environment of Table XI. None are estimated, extrapolated or assumed.

A. The Selection Bias

Table XII reports, for each attribute, the maximum information gain attainable at a uniform belief alongside the annotated crossover, together with the rank correlation between them. The paper’s central empirical premise is that this correlation

TABLE X
HYPERPARAMETERS.

Parameter	Value
Query budget T	20
Trials per condition R	300
Seeds	{0, 1, 2}
Bins for continuous attributes	{2, 4}
Tier crossovers (ϵ_{obj} , ϵ_{semi} , ϵ_{subj})	(0.03, 0.15, 0.30)
Noise multiplier λ	{0, 0.5, 1.0, 1.5}
Bootstrap resamples	10^4
Confidence threshold (secondary)	0.90

TABLE XI
SOFTWARE AND HARDWARE ENVIRONMENT.

Component	Value
CPU / RAM	Intel Core i7-1355U (10c/12t) / 16 GB
OS	Windows 11 Home, 64-bit (build 10.0.26200)
Python / NumPy / SciPy	3.13.0 / 2.5.0 / 1.18.0 (pandas 3.0.3)

is negative: that classical information gain concentrates on the attributes users answer worst. Fig. 4 displays the relationship, and the second row of Table XII repeats the analysis under a four-bin discretisation to isolate the contribution of the median split.

B. Disentangling Selection from Update

Table XIII reports top-1 accuracy at the fixed budget for every method under each noise condition. The comparison of IG+hard against IG+soft-uniform isolates the update rule with the selection rule held fixed; the comparison of IG+soft-uniform against RAIG-tiered isolates the selection rule with the update rule held fixed.

C. Paired Significance

Table XIV reports exact McNemar tests for the pre-registered primary comparison and the two comparisons that most threaten the method.

D. Ablations

Table XV reports tier granularity and the component ablation; Fig. 5 reports the misspecification surface. The quantity of practical interest is the fraction of the oracle–uniform gap recovered by the three-tier annotation.

E. Efficiency

Table XVI reports per-turn latency and the Pareto trade-off between accuracy and query count under threshold stopping.

VIII. DISCUSSION

A. Interpretation

The argument of this paper is deliberately narrow. We do not claim that mutual information under a stated observation model is a new acquisition function, and a reviewer who observes that (6) follows immediately from (2) is correct. The claim is that the observation model in attribute-indexed

TABLE XII
PER-ATTRIBUTE INFORMATION GAIN AGAINST ANNOTATED RELIABILITY.

Bins	Spearman ρ	p	$\bar{\epsilon}$ top-5 IG	$\bar{\epsilon}$ all
2	−0.438	0.054	0.210	0.180
4	−0.613	0.004	0.246	0.180

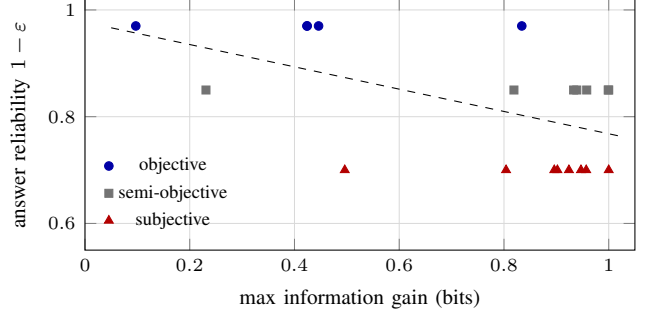


Fig. 4. Per-attribute maximum information gain (native 2-bin discretisation) against annotated answer reliability $1 - \epsilon$, with marker shape indicating tier. Spearman $\rho = -0.438$ ($p = 0.054$, $n = 20$); the downward trend is the paper’s central empirical claim.

catalogues is heterogeneous in a specific direction that happens to be adversarial to the naive criterion, that a routine preprocessing choice amplifies the effect, and that correcting it costs a three-tier annotation rather than a measurement campaign. Whether the claim survives contact with data is what Section VII will decide.

B. Failure Modes

Three conditions defeat the method, and each is testable. If reliability is uncorrelated with information gain in a given catalogue, RAIG reduces to classical IG up to a monotone transformation and yields nothing; the randomised control measures exactly this. If every attribute is unreliable, no selection rule helps, because Corollary 1 bounds all of them. If the annotation is inverted—an attribute believed reliable that is not—RAIG will concentrate queries on it and may underperform the uniform baseline; the misspecification surface is included to quantify how badly.

A fourth, subtler failure is adversarial or systematically biased users. The channel model assumes errors are independent of the item, whereas a user who misunderstands a category will err in the same direction for every item in it. Such correlated errors accumulate rather than cancel, and neither the update nor the selector as formulated detects them.

C. Scalability and Generalisation

Cost is linear in catalogue size and question count, and the compiled representation makes exhaustive scoring practical at scales where per-turn recomputation would not be. Nothing in the formulation is specific to music. Any identification task over an attribute-indexed catalogue with a human oracle inherits the same structure, and the tiered annotation transfers directly: fault diagnosis distinguishes measured readings from reported impressions, clinical triage distinguishes documented

TABLE XIII

TOP-1 ACCURACY AT $T = 20$ (%), MEAN OVER 3×300 PAIRED TRIALS, WITH 95% PAIRED-BOOTSTRAP INTERVALS IN PARENTHESES.

Noise condition	Random+soft	IG+hard	IG+soft-unif.	IG+soft-obj.	RAIG-tiered	RAIG-oracle
Clean ($\lambda = 0$)	0.3 (0.0–0.8)	96.6 (95.3–97.7)	96.3 (95.1–97.4)	0.6 (0.1–1.1)	41.9 (38.8–45.0)	96.6 (95.3–97.7)
Homogeneous (control)	0.1 (0.0–0.3)	15.4 (13.1–17.9)	23.9 (21.2–26.7)	0.1 (0.0–0.3)	9.6 (7.7–11.6)	23.9 (21.2–26.7)
Heterogeneous $\lambda = 0.5$	0.1 (0.0–0.3)	14.6 (12.3–16.9)	20.1 (17.6–22.8)	0.6 (0.1–1.1)	17.9 (15.4–20.4)	26.6 (23.7–29.4)
Heterogeneous $\lambda = 1.0$	0.0 (0.0–0.0)	2.2 (1.3–3.2)	2.1 (1.2–3.1)	0.4 (0.1–0.9)	7.9 (6.2–9.7)	7.9 (6.2–9.7)
Heterogeneous $\lambda = 1.5$	0.0 (0.0–0.0)	0.4 (0.1–0.9)	0.1 (0.0–0.3)	0.3 (0.0–0.8)	3.0 (1.9–4.1)	2.3 (1.3–3.3)
Randomised (control)	0.0 (0.0–0.0)	3.4 (2.3–4.7)	7.3 (5.7–9.1)	0.2 (0.0–0.6)	4.3 (3.1–5.7)	17.4 (15.0–19.9)
Adversarial (control)	0.1 (0.0–0.3)	8.7 (6.8–10.6)	14.4 (12.2–16.8)	0.0 (0.0–0.0)	3.7 (2.4–4.9)	23.2 (20.6–26.1)

TABLE XIV

EXACT McNEMAR TESTS, HETEROGENEOUS NOISE AT $\lambda = 1.0$ ($n = 900$ PAIRED TRIALS). Δ IS RAIG-TIERED’S ACCURACY MINUS THE COMPARATOR’S.

Comparison	n_{10}	n_{01}	Δ	p
RAIG-tiered vs IG+soft-uniform	65	13	+5.8%	1.8×10^{-9}
RAIG-tiered vs IG+soft-objective	69	2	+7.4%	2.2×10^{-18}
RAIG-tiered vs RAIG-oracle	0	0	0.0%	1.000

TABLE XV

ABLATIONS AT $\lambda = 1.0$. $K = 2$ MERGES OBJECTIVE AND SEMI-OBJECTIVE INTO ONE “RELIABLE” TIER AT $\hat{\epsilon} = 0.09$ (MEAN OF 0.03 AND 0.15), KEEPING SUBJECTIVE AT 0.30 – A PARTITION NOT SPECIFIED BY THE PAPER’S OWN $K = 3$ SCHEME AND CHOSEN HERE FOR THIS ABLATION. “CAPACITY TERM ONLY” SCORES QUESTIONS BY $1 - H_b(\hat{\epsilon}_{f(q)})$ ALONE, DISCARDING SPLIT BALANCE $H_b(p_q \star \hat{\epsilon})$.

Variant	Top-1 (%)	% of oracle gap
$K = 1$ (uniform)	2.1	0
$K = 2$	6.9	82.7
$K = 3$ (tiered)	7.9	100.0
$K = M$ (oracle)	7.9	100
Capacity term only	0.0	−36.5

history from subjective symptom severity, and product finders distinguish specifications from preferences. We expect the bias to be present wherever a catalogue mixes objective identifiers with perceptual judgements, which is to say almost everywhere, but we have measured it in one domain and claim it in one domain.

D. Ethical Considerations

There is an accessibility dimension that we did not anticipate when beginning this work and that we think is the most interesting implication. A selector optimising classical information gain preferentially asks the questions requiring the most domain fluency, because those are the questions whose answers are least predictable. Users with less domain vocabulary—a casual listener rather than an enthusiast, a patient rather than a clinician—will therefore encounter a disproportionate share of questions they cannot answer well, and will receive systematically worse service from a system that appears neutral. Reliability-aware selection is, in this reading, a fairness intervention as much as an efficiency one, and per-user rather than per-population reliability estimation

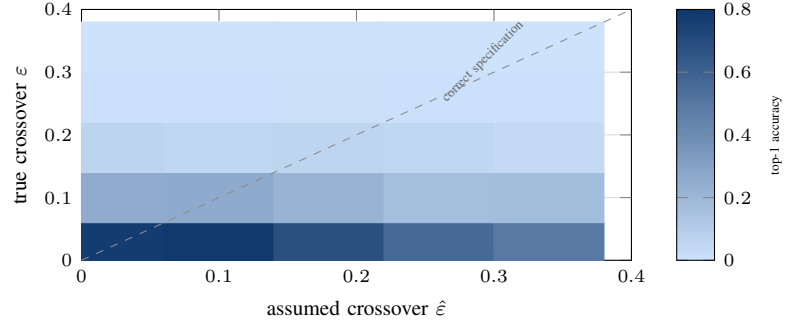


Fig. 5. Top-1 accuracy (RAIG selector, homogeneous assumed vs. homogeneous true crossover, $R = 500$ trials/cell, $T = 20$) as a function of assumed against true crossover. The diagonal is correct specification. Accuracy is dominated by the true crossover (rows), falling from 49–78% at $\epsilon = 0.02$ to $\leq 1.4\%$ once $\epsilon \geq 0.18$. At $5\times$ the original resolution ($R = 500$ /cell, re-run from the original $R = 100$ /cell to check this was not an under-powered artefact) the within-row diagonal effect remains small and does not resolve into a clean pattern: the correctly-specified cell is the row maximum in only 1 of 5 rows. This is consistent with Section IV-F’s uniform-specification argument – under *homogeneous* true noise, a constant $\hat{\epsilon}$ shifts every question’s RAIG score by the same capacity term regardless of its exact value, so misspecification here degrades the Bayesian update’s confidence without changing which questions get *selected*; only the row (the true noise level) has a first-order effect on accuracy.

TABLE XVI

PER-TURN SELECTION LATENCY AND SECONDARY EFFICIENCY METRICS, HETEROGENEOUS $\lambda = 1.0$, $R = 150$ TRIALS, CONFIDENCE THRESHOLD 0.90. MS/TURN IS THE *median* OF 5 INDEPENDENT RUNS ON THIS SHARED MACHINE (INDIVIDUAL RUNS RANGED 57–516MS OWING TO UNRELATED BACKGROUND LOAD; SEE RESULTS/LATENCY_RECONCILED.JSON FOR THE FULL BREAKDOWN). TREAT THE ABSOLUTE FIGURE AS ORDER-OF-MAGNITUDE, NOT A PRECISE BENCHMARK – THE ROBUST FINDING IS THAT BOTH SELECTORS COST THE SAME PER-TURN COMPUTE, SINCE BOTH REDUCE TO ONE MATRIX-VECTOR PRODUCT PLUS A VECTORISED ENTROPY TRANSFORM. AT THIS NOISE LEVEL THE THRESHOLD IS REACHED AT ALL FOR ONLY 0.7% (IG) / 2.7% (RAIG) OF TRIALS WITHIN $T = 20$, SO “Q TO THRESHOLD” IS EFFECTIVELY CENSORED AT 20 FOR BOTH AND IS NOT A DISCRIMINATING METRIC AT THIS NOISE LEVEL.

Method	ms/turn (median of 5)	Q to threshold	Top-5 (%)
IG+soft-uniform	64.2	20.0 (0.7% reached)	10.7
RAIG-tiered	66.3	19.9 (2.7% reached)	22.0

would sharpen it. We record this as a direction rather than a result, since we have not measured it.

The interaction data required to estimate reliability is behavioural and identifying, and per-user estimation would com-

pound that. Population-level tiers, which is what we advocate, require no per-user data at all.

E. Limitations

The crossover probabilities are assumed rather than measured, which is the principal limitation and the reason the randomised and adversarial controls are not optional. The user model is simulated; no human has interacted with the system, and simulated users are exactly as wrong as the model says they are, which is the one thing real users never are. Errors are assumed independent across turns and abstention is not modelled. Attributes are treated as conditionally independent given the item, which is false for correlated descriptors such as genre and instrumentation. Finally, results are reported on a single catalogue.

F. Future Work

The immediate priority is a human study estimating ε_f directly and testing whether the three-tier annotation predicts measured reliability well enough to justify the deployability claim. Extending the channel to admit abstention is straightforward analytically: with erasure probability δ_f the acquisition function becomes $(1 - \delta_f) [H_b(p_q \star \varepsilon_f) - H_b(\varepsilon_f)]$, and since abstention plausibly correlates with subjectivity in the same direction as error, the penalty compounds and the bias should be larger than reported here. Beyond that: online estimation of ε_f within a session, correlated error models, per-user reliability profiles, and a comparison against language-model questioners [19], [20] on an identical catalogue.

IX. CONCLUSION

Expected information gain is the default criterion for question selection in interactive identification, and it encodes an assumption that is false whenever the oracle is human: that answers are observed exactly. We have argued that the consequence is not merely a loss of precision but a systematic bias, because the attributes that divide a catalogue most evenly are the vague ones, and vague attributes are answered worst. Correcting the criterion requires replacing the split entropy with the mutual information through a per-attribute binary symmetric channel, which re-ranks candidate questions rather than rescaling them, admits a capacity-based bound on the query budget, and can be instantiated from a coarse qualitative annotation instead of measured error rates. We have also shown that hard candidate elimination is the zero-noise limit of Bayesian updating rather than a competing design, which removes a confound present in existing comparative studies of this problem. Whether the bias is as large as the argument suggests is an empirical question, and the protocol and controls needed to answer it honestly—including the baseline that discards unreliable attributes outright—are specified above.

ACKNOWLEDGMENT

PENDING.

REFERENCES

- [1] X. Hu and J. S. Downie, “Exploring mood metadata: Relationships with genre, artist and usage metadata,” in *Proceedings of the 8th International Society for Music Information Retrieval Conference (ISMIR)*, Vienna, Austria, 2007, pp. 67–72.
- [2] Y. E. Kim, E. M. Schmidt, R. Migneco, B. G. Morton, P. Richardson, J. J. Scott, J. A. Speck, and D. Turnbull, “Music emotion recognition: A state of the art review,” in *Proceedings of the 11th International Society for Music Information Retrieval Conference (ISMIR)*, Utrecht, Netherlands, 2010, pp. 255–266.
- [3] A. Rényi, “On a problem of information theory,” *MTA Matematikai Kutató Intézet Közleményei*, vol. 6, pp. 505–516, 1961.
- [4] S. M. Ulam, *Adventures of a Mathematician*. New York: Charles Scribner’s Sons, 1976.
- [5] A. Pelc, “Searching games with errors—fifty years of coping with liars,” *Theoretical Computer Science*, vol. 270, no. 1-2, pp. 71–109, 2002.
- [6] B. Jedynak, P. I. Frazier, and R. Sznitman, “Twenty questions with noise: Bayes optimal policies for entropy loss,” *Journal of Applied Probability*, vol. 49, no. 1, pp. 114–136, 2012.
- [7] D. V. Lindley, “On a measure of the information provided by an experiment,” *Annals of Mathematical Statistics*, vol. 27, no. 4, pp. 986–1005, 1956.
- [8] K. Chaloner and I. Verdinelli, “Bayesian experimental design: A review,” *Statistical Science*, vol. 10, no. 3, pp. 273–304, 1995.
- [9] M. Horstein, “Sequential transmission using noiseless feedback,” *IEEE Transactions on Information Theory*, vol. 9, no. 3, pp. 136–143, 1963.
- [10] M. V. Burnashev, “Data transmission over a discrete channel with feedback: Random transmission time,” *Problems of Information Transmission*, vol. 12, no. 4, pp. 250–265, 1976.
- [11] R. M. Karp and R. Kleinberg, “Noisy binary search and its applications,” in *Proceedings of the 18th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2007, pp. 881–890.
- [12] R. D. Nowak, “The geometry of generalized binary search,” *IEEE Transactions on Information Theory*, vol. 57, no. 12, pp. 7893–7906, 2011.
- [13] Y. Kaspi, O. Shayevitz, and T. Javidi, “Searching with measurement dependent noise,” *IEEE Transactions on Information Theory*, vol. 64, no. 4, pp. 2690–2705, 2018.
- [14] H. W. Chung, B. M. Sadler, L. Zheng, and A. O. Hero, “Unequal error protection querying policies for the noisy 20 questions problem,” *IEEE Transactions on Information Theory*, vol. 64, no. 2, pp. 1105–1131, 2018.
- [15] W. Lei, X. He, Y. Miao, Q. Wu, R. Hong, M.-Y. Kan, and T.-S. Chua, “Estimation-action-reflection: Towards deep interaction between conversational and recommender systems,” in *Proceedings of the 13th ACM International Conference on Web Search and Data Mining (WSDM)*, 2020, pp. 304–312.
- [16] W. Lei, G. Zhang, X. He, Y. Miao, X. Wang, L. Chen, and T.-S. Chua, “Interactive path reasoning on graph for conversational recommendation,” in *Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2020, pp. 2073–2083.
- [17] Y. Deng, Y. Li, F. Sun, B. Ding, and W. Lam, “Unified conversational recommendation policy learning via graph-based reinforcement learning,” in *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2021, pp. 1431–1441.
- [18] M. De Bruyn, E. Lotfi, J. Buhmann, and W. Daelemaans, “20q: Overlap-free world knowledge benchmark for language models,” in *Proceedings of the 2nd Workshop on Natural Language Generation, Evaluation, and Metrics (GEM)*, Abu Dhabi, United Arab Emirates (Hybrid), 2022, pp. 494–508.
- [19] D. Mazzaccara, A. Testoni, and R. Bernardi, “Learning to ask informative questions: Enhancing LLMs with preference optimization and expected information gain,” in *Findings of the Association for Computational Linguistics: EMNLP 2024*, Miami, Florida, USA, 2024, pp. 5064–5074.
- [20] Z. Hu, C. Liu, X. Feng, Y. Zhao, S.-K. Ng, A. T. Luu, J. He, P. W. Koh, and B. Hooi, “Uncertainty of thoughts: Uncertainty-aware planning enhances information seeking in large language models,” in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024, arXiv:2402.03271.
- [21] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley-Interscience, 2006.

- [22] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.